UME

Organization / Business Operating System

Users, Workflows & Use Cases

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1. User Personas

UME serves a broad range of organizational stakeholders. The personas below define the primary user types, their objectives, the modules they interact with most, and the challenges the UME OS is designed to solve for them.

Executive & Leadership Personas

Persona	P-01 — Chief Executive Officer (CEO)
Level	C-Suite / Executive
Description	Leads the entire organization; requires a consolidated real-time view of organizational health, strategic progress, risk posture, and financial performance across all functions.
Key Modules	Strategy (21), Analytics (02), Risk (33), Board (04), Portal (26), Finance (14)
Goals	Monitor OKR progress; review top risks; approve strategic initiatives; board preparation
Pain Points	Data fragmented across systems; no single source of truth; slow executive reporting cycles

Persona	P-02 — Chief Financial Officer (CFO)
Level	C-Suite / Executive
Description	Owns financial health, capital allocation, investor reporting, and fiscal governance. Needs real-time visibility across all financial domains including treasury, budget, and compliance.
Key Modules	Finance (14), Investment (17), GRC (15), Analytics (02), Board (04), Risk (33)
Goals	Close the financial books; manage budget variance; approve capital allocation; produce investor reports
Pain Points	Slow month-end consolidation; manual variance analysis; disconnected budget vs actuals

Persona	P-03 — Chief Operating Officer (COO)
Level	C-Suite / Executive
Description	Accountable for operational efficiency, service delivery, and cross-functional process performance. Needs dashboards showing throughput, SLA compliance, and operational exceptions.
Key Modules	Operations (25), Process (29), Analytics (02), Supply Chain (37), Portal (26)
Goals	Optimize operational performance; enforce SLAs; manage cross-functional bottlenecks
Pain Points	No unified view of operational KPIs; process bottlenecks invisible until escalated

Persona	P-04 — Chief People Officer (CHRO)
Level	C-Suite / Executive
Description	Responsible for talent strategy, workforce planning, culture, and total rewards. Needs visibility across recruiting, retention, learning, and performance.
Key Modules	HR (16), Learning (20), Analytics (02), Teams (38), Work (40)

Goals	Reduce regrettable attrition; build capability; monitor workforce diversity; drive engagement
Pain Points	People data siloed in disconnected HRIS; limited predictive workforce analytics

Persona	P-05 — Chief Technology Officer (CTO)
Level	C-Suite / Executive
Description	Leads technology strategy, engineering productivity, and architectural governance. Needs oversight of tech stack health, technical debt, and engineering delivery metrics.
Key Modules	Engineering (10), IT (18), Security (36), Product (30), Analytics (02)
Goals	Drive engineering velocity; reduce tech debt; govern architecture decisions; manage EOL risk
Pain Points	Technical debt invisible to leadership; fragmented incident and change visibility

Management & Operations Personas

Persona	P-06 — Department Manager
Level	Senior Manager / Director
Description	Manages a functional department (e.g., Finance, Marketing, HR). Responsible for team performance, budget adherence, and delivery against departmental OKRs.
Key Modules	Work (40), Analytics (02), HR (16), Finance (14), Teams (38), Portal (26)
Goals	Keep team on track; manage budget; report departmental performance; resolve blockers
Pain Points	Too much time in status meetings; limited visibility into team workload and blockers

Persona	P-07 — Project Manager
Level	Manager / Lead
Description	Owns delivery of one or more projects within the portfolio. Plans, schedules, tracks, and reports on project scope, cost, quality, risks, and stakeholder communications.
Key Modules	Portfolio (27), Work (40), Risk (33), Schedule (35), Process (29)
Goals	Deliver projects on time and on budget; manage risks; communicate status to stakeholders
Pain Points	Scattered project data across tools; manual status reporting; RAID log maintenance burden

Persona	P-08 — Operations Manager
Level	Manager
Description	Manages day-to-day operations of a unit or function. Monitors SLAs, manages work queues, resolves escalations, and drives continuous improvement.
Key Modules	Operations (25), Process (29), Work (40), Analytics (02), Schedule (35)
Goals	Maintain SLA compliance; manage team capacity; identify and remove bottlenecks
Pain Points	SLA breaches discovered reactively; limited process performance visibility

Persona	P-09 — Finance Manager
Level	Manager / Specialist
Description	Manages accounting, financial reporting, and budget administration for a business unit or the whole organization.
Key Modules	Finance (14), Analytics (02), GRC (15), Supply Chain (37)
Goals	Accurate financial statements; budget adherence; timely AP/AR; clean audit trail
Pain Points	Manual reconciliation; spreadsheet-based budgeting; slow close cycles

Persona	P-10 — HR Manager
Level	Manager / Specialist
Description	Manages employment lifecycle, talent acquisition, performance cycles, and employee relations for a business unit.
Key Modules	HR (16), Learning (20), Work (40), Analytics (02), Communications (07)
Goals	Fill vacancies quickly; run fair performance cycles; maintain compliance; engage employees
Pain Points	Candidate tracking in email; manual onboarding checklists; disconnected L&D tracking

Specialist & Contributor Personas

Persona	P-11 — Legal & Compliance Officer
Level	Specialist
Description	Responsible for legal entity management, regulatory compliance, contract governance, and legal risk management.
Key Modules	Legal Entity (13), GRC (15), CMS (06), Board (04), Risk (33)
Goals	Maintain entity compliance; track filing deadlines; manage contracts; assess regulatory changes
Pain Points	Filing deadlines tracked in spreadsheets; compliance status invisible until audit

Persona	P-12 — Risk Manager
Level	Specialist
Description	Owns the enterprise risk framework, maintains the risk register, monitors KRIs, and coordinates treatment plans.
Key Modules	Risk (33), GRC (15), Analytics (02), Board (04)
Goals	Current risk register; proactive KRI monitoring; clear treatment ownership; board reporting
Pain Points	Risk data stale between review cycles; KRI thresholds not automatically monitored

Persona	P-13 — Sales Representative
Level	Contributor

Description	Manages a pipeline of opportunities, builds client relationships, produces proposals, and closes business.
Key Modules	Sales (34), CRM (08), Product (30), Communications (07), Schedule (35)
Goals	Hit quota; manage pipeline hygiene; log activities; collaborate with marketing on leads
Pain Points	CRM data entry burden; no visibility into marketing-generated leads; manual forecasting

Persona	P-14 — Marketing Manager
Level	Specialist
Description	Plans and executes marketing campaigns, manages channel budgets, tracks campaign performance, and drives lead generation.
Key Modules	Marketing (22), CRM (08), Analytics (02), Communications (07), Design (09)
Goals	Launch campaigns on schedule; optimize channel mix; measure ROI; generate quality leads
Pain Points	Campaign data fragmented; signal-to-insight pipeline manual; creative approval slow

Persona	P-15 — Software Engineer
Level	Contributor
Description	Develops and maintains software products. Manages work items, participates in planning, tracks technical debt, and contributes to architecture decisions.
Key Modules	Work (40), Engineering (10), Product (30), Requirements (32), IT (18)
Goals	Clear requirements; unblocked work items; visible tech debt; low MTTR on incidents
Pain Points	Requirements ambiguity; siloed technical decisions; context-switching from fire-fighting

Persona	P-16 — IT Administrator
Level	Specialist
Description	Manages IT assets, infrastructure, incidents, changes, and service requests. Maintains the CMDB and governs license compliance.
Key Modules	IT (18), Security (36), Work (40), Analytics (02)
Goals	Zero unplanned downtime; patch all critical CVEs; license compliance; fast incident resolution
Pain Points	Asset discovery manual; change control paper-based; license sprawl invisible

Persona	P-17 — Knowledge Worker / Employee
Level	Contributor
Description	Day-to-day employee who consumes org services, manages their own work items, accesses knowledge, takes learning courses, and interacts with the organization through the portal.
Key Modules	Portal (26), Work (40), Knowledge (19), Learning (20), HR (16), Communications (07)
Goals	Find information quickly; complete work tasks; access HR services; stay informed

Pain Points	Information scattered; no single inbox for notifications; learning content hard to discover

Persona	P-18 — UME System Operator / Admin
Level	System Administrator
Description	Manages the UME platform itself: module configuration, RBAC administration, system health monitoring, backup scheduling, and user provisioning.
Key Modules	Enterprise Mgmt (12), Security (36), Backup (03), IT (18), Admin (01)
Goals	System always available; configuration changes tracked; fast incident response; clean audit log
Pain Points	No visibility into module health; config changes ad-hoc; backup verification manual

2. Cross-Cutting & Enterprise Workflows

These workflows span multiple modules and user types. They represent the highest-value end-to-end organizational journeys in the UME OS.

WF-01: New Employee Onboarding

A new hire joins the organization. The UME OS orchestrates the complete onboarding journey from job offer acceptance through system provisioning, orientation, and first-90-days milestone tracking.

#	Actor	Action	Module	Output / Next State
1	**HR Manager (P-10)**	Create employee record and set employment status to Pre-Start	*HR Module (16)*	Employee record created; org unit assigned
2	**HR Module (automated)**	Trigger onboarding workflow via Process Module	*Process Module (29)*	Onboarding workflow instance started; task list generated
3	**IT Administrator (P-16)**	Receive IT provisioning task; create accounts, assign hardware CI	*IT Module (18)*	Accounts provisioned; CMDB updated with assigned assets
4	**Facilities Manager**	Receive desk/access badge provisioning task; assign workspace	*Facilities Module (24)*	Workspace allocated; badge access record created
5	**HR Module (automated)**	Enroll employee in mandatory compliance training	*Learning Module (20)*	Compliance training enrollments created with due dates
6	**HR Module (automated)**	Distribute employee welcome communications package	*Communications Module (07)*	Welcome email, org chart invite, handbook sent
7	**HR Manager (P-10)**	Assign 90-day development plan and initial OKRs	*Work Module (40) + HR (16)*	Work items created; learning path assigned
8	**Manager (P-06)**	Schedule 30/60/90 check-in meetings	*Schedule Module (35)*	Recurring check-in events on enterprise calendar
9	**Employee (P-17)**	Access portal; complete profile; begin orientation content	*Portal (26) + Knowledge (19)*	Portal session logged; orientation article marked read
10	**HR Module (automated)**	Monitor compliance training completion; escalate if overdue	*Learning Module (20)*	Training status updated; overdue alert if breached

Events Emitted

hr.employee.hired | process.workflow.started | it.asset.assigned
learning.compliance.enrolled | comms.message.sent | schedule.event.created
learning.content.completed (on each orientation module) | hr.onboarding.complete

WF-02: Monthly Financial Close

The organization runs its monthly financial close cycle. This workflow spans finance, legal entities, analytics, and executive reporting.

#	Actor	Action	Module	Output / Next State
1	**Finance Manager (P-09)**	Initiate month-end close workflow via Process Module	*Process Module (29)*	Close checklist workflow started; task owners notified
2	**Finance Manager (P-09)**	Post final journal entries for the period	*Finance Module (14)*	Journal entries posted; period locked from further posting
3	**Finance Module (automated)**	Run AP/AR aging and match open invoices	*Finance Module (14)*	Aged AP/AR report; unmatched items escalated
4	**Finance Module (automated)**	Compute budget vs actuals variance analysis	*Finance Module (14)*	Variance report; alerts for threshold breaches
5	**Finance Manager (P-09)**	Reconcile multi-entity intercompany eliminations	*Finance Module (14) + Legal Entity (13)*	Intercompany eliminations applied; consolidated P&L draft
6	**Finance Module (automated)**	Generate financial statements for each entity	*Finance Module (14)*	P&L, Balance Sheet, Cash Flow Statement per entity
7	**Analytics Module (automated)**	Refresh finance KPI dashboards with period-end data	*Analytics Module (02)*	CFO and board dashboards updated
8	**CFO (P-02)**	Review statements; sign off on period close	*Finance Module (14)*	Period formally closed; sign-off audit record created
9	**Finance Module (automated)**	Distribute financial reports to board and executive team	*Communications Module (07) + Templates (39)*	Branded financial pack distributed to recipients

Events Emitted

finance.journal.posted | finance.period.locked | finance.budget.variance_exceeded (if triggered)
finance.statement.generated | analytics.dashboard.refreshed | finance.period.closed

WF-03: Annual Strategic Planning Cycle

The organization runs its annual strategy refresh, setting direction for the coming year through a structured sequence of analysis, planning, OKR setting, and board approval.

#	Actor	Action	Module	Output / Next State
1	**CEO (P-01) + Strategy Team**	Commission environmental scanning: PESTLE and SWOT analysis	*Strategy Module (21)*	PESTLE and SWOT analyses recorded; strategic implications captured
2	**Strategy Team**	Review business development pipeline and market intelligence	*Biz Dev Module (05)*	Pipeline insights and growth opportunities incorporated
3	**CEO + Department Heads**	Facilitate strategic planning workshop; define pillars and 3-year objectives	*Strategy Module (21)*	Strategic objectives created and owned; horizon plan defined

4	**Department Managers (P-06)**	Draft departmental OKRs cascaded from strategic objectives	*Strategy Module (21)*	OKRs created at department level; targets set for period
5	**CFO (P-02)**	Validate OKRs against financial plan; link to budget lines	*Finance Module (14) + Strategy (21)*	Budget lines linked to strategic initiatives; financial viability confirmed
6	**Risk Manager (P-12)**	Assess strategic risks associated with new objectives	*Risk Module (33)*	Strategic risks registered; KRIs defined for new objectives
7	**Strategy Team**	Assemble strategy deck and board presentation pack	*Board Module (04) + Templates (39)*	Strategy board pack assembled; agenda item added
8	**Board of Directors**	Review and approve strategic plan; pass resolution	*Board Module (04)*	Strategy formally approved; resolution recorded
9	**Strategy Module (automated)**	Publish OKRs to all modules; activate KPI targets	*Strategy Module (21) + Analytics (02)*	OKRs distributed; KPI dashboards activated with new targets
10	**All Managers (P-06)**	Communicate strategy to teams; assign team-level OKRs	*Communications (07) + Teams (38)*	Team OKRs set; staff communications distributed

WF-04: Security Incident Response

A security incident is detected. The UME OS coordinates triage, containment, investigation, remediation, and post-incident review.

#	Actor	Action	Module	Output / Next State
1	**Security Module (automated)**	Detect anomaly via SIEM integration; create incident record	*Security Module (36)*	SecurityIncident created; severity classified; P1 timer starts
2	**Security Module (automated)**	Activate crisis communication playbook; notify response team	*Communications Module (07)*	Response team notified within SLA; incident bridge opened
3	**IT Admin (P-16)**	Identify affected CIs via CMDB impact analysis	*IT Module (18)*	Affected configuration items and systems listed
4	**Security Team**	Contain incident: isolate affected systems via change requests	*IT Module (18) + Security (36)*	Change requests raised and approved; containment applied
5	**Security Team**	Investigate root cause; document findings	*Security Module (36)*	Root cause documented; attack vector identified
6	**Security Module (automated)**	Assess if personal data affected; trigger privacy breach notification workflow	*Security (36) + Process (29)*	DSAR/breach notification workflow started if required
7	**IT Admin (P-16)**	Apply patches and remediate vulnerability	*Security (36) + IT (18)*	Vulnerability record closed; patch applied; CMDB updated
8	**Security Team**	Close incident; write Post-Incident Review (PIR)	*Security Module (36)*	Incident closed; PIR document attached; lessons logged

9	**Knowledge Manager**	Capture lessons learned in knowledge base	*Knowledge Module (19)*	Lessons learned article published; related runbook updated
10	**Risk Manager (P-12)**	Update risk register based on PIR findings	*Risk Module (33)*	Risk record updated or new risk created; KRI thresholds reviewed

Events Emitted

security.incident.opened | comms.crisis.activated | it.change.approved
security.privacy.breach_detected (if triggered) | security.incident.resolved
knowledge.lesson.captured | risk.assessment.completed

WF-05: New Product Launch

A new product or service is brought to market. This cross-functional workflow coordinates engineering, marketing, sales, legal, and operations.

#	Actor	Action	Module	Output / Next State
1	**Product Manager**	Create product record; define roadmap and initial feature backlog	*Product Module (30)*	Product catalog entry created; backlog populated
2	**Business Analyst**	Capture and baseline business requirements	*Requirements Module (32)*	Requirements created; stakeholder sign-off obtained
3	**Legal Officer (P-11)**	Review product for regulatory compliance; check entity requirements	*Legal Entity Module (13) + GRC (15)*	Compliance checklist passed; any entity changes identified
4	**Engineering Team (P-15)**	Build product features; log ADRs; track tech debt	*Engineering (10) + Work (40)*	Features delivered; ADRs recorded; sprint velocity tracked
5	**Product Manager**	Plan and approve product release; run go/no-go gate	*Product Module (30)*	Release plan confirmed; go-no-go decision recorded
6	**Marketing Manager (P-14)**	Register launch campaign; plan audience and channel strategy	*Marketing Module (22)*	Campaign registered; strategy packs evaluated; insights generated
7	**Finance Manager (P-09)**	Set product pricing; publish price list	*Product (30) + Finance (14)*	Price list published; revenue recognition rules configured
8	**Sales Rep (P-13)**	Complete product training; add to CRM opportunity lines	*Learning (20) + Sales (34)*	Sales enablement complete; product available in quote tool
9	**Marketing Module (automated)**	Execute launch campaign; dispatch communications	*Marketing (22) + Communications (07)*	Campaign live; press release distributed; emails sent
10	**Analytics Module (automated)**	Begin tracking product KPIs: adoption, revenue, NPS	*Analytics (02) + Product (30)*	Product performance dashboard live; KPI baselines established

WF-06: End-to-End Procurement Cycle

A business unit needs to procure goods or services. The workflow covers requisition, approval, PO issuance, receiving, and payment.

#	Actor	Action	Module	Output / Next State
1	**Requester (P-17)**	Submit purchase requisition via work management	*Work Module (40)*	Requisition work item created; pending approval
2	**Manager (P-06)**	Approve requisition; validate budget availability	*Work (40) + Finance (14)*	Requisition approved; budget check passed
3	**Procurement Team**	Issue RFQ to approved suppliers; evaluate responses	*Supply Chain Module (37)*	RFQ sent to approved vendor list; responses collected
4	**Procurement Team**	Select supplier; create and approve purchase order	*Supply Chain Module (37)*	PO created and issued to supplier; delivery date confirmed
5	**Warehouse / Receiving**	Receive goods; complete goods receipt against PO lines	*Supply Chain Module (37)*	GR recorded; inventory updated; quality inspection triggered
6	**Finance Module (automated)**	Run 3-way match: PO vs GR vs supplier invoice	*Finance Module (14)*	Invoice matched; approved for payment or variance escalated
7	**Finance Manager (P-09)**	Approve payment; post payment journal entry	*Finance Module (14)*	Supplier invoice paid; journal posted; audit record created
8	**Analytics Module (automated)**	Update procurement KPIs: savings, cycle time, OTIF	*Analytics (02)*	Procurement dashboard refreshed

WF-07: Internal Compliance Audit

The GRC team conducts a compliance audit engagement covering a selected regulatory domain.

#	Actor	Action	Module	Output / Next State
1	**Internal Auditor**	Create audit engagement; define scope and assign auditors	*GRC Module (15)*	Audit engagement created; scope documented; team notified
2	**GRC Module (automated)**	Pull relevant compliance obligations and associated controls	*GRC Module (15)*	Obligation and control list pre-populated for scope
3	**Internal Auditor**	Request evidence from control owners via process workflow	*Process Module (29) + Work (40)*	Evidence request tasks sent to owners; due dates set
4	**Control Owners**	Upload evidence and self-assessment responses	*GRC (15) + CMS (06)*	Evidence uploaded; control self-assessments recorded
5	**Internal Auditor**	Test controls; document findings and exceptions	*GRC Module (15)*	Control test results recorded; findings raised
6	**GRC Module (automated)**	Escalate critical findings to Risk Manager and COO	*GRC (15) + Communications (07)*	Critical findings communicated; escalation audit record created

7	**Internal Auditor**	Issue draft audit report; request management response	*GRC (15) + Templates (39)*	Draft report generated from template; management responses tracked
8	**Management**	Agree remediation plans with owners and due dates	*Work Module (40)*	Remediation work items created; owners assigned; dates confirmed
9	**Internal Auditor**	Finalize and publish audit report	*GRC Module (15) + CMS (06)*	Final report published; distributed to CEO, CFO, Board Audit Committee
10	**GRC Module (automated)**	Track remediation progress; escalate overdue items	*GRC (15) + Work (40)*	Remediation status monitored; overdue alerts emitted

3. Domain-Specific Use Cases

This section details individual use cases organized by organizational domain. Each use case specifies actors, triggers, main flow, alternatives, and postconditions.

3.1 Administration

UC-ADM-01 Register a New Organizational Unit

Actors	Trigger	Modules Involved
HR Manager (P-10), Department Manager (P-06)	Business restructure or new team approved by executive leadership	Administration (01), HR (16), Finance (14), Analytics (02)

Preconditions

- Parent org unit exists in hierarchy
- Manager employee record exists in HR module
- Budget code assigned by Finance

Main Success Flow

1. Manager submits org unit registration request through the portal.
2. HR Manager opens Administration Module and creates OrgUnit record with name, type, parent, and cost center.
3. Administration Module validates parent_id exists and manager_id resolves to active employee.
4. HR Module is notified: updates headcount reporting and links employees to new unit.
5. Finance Module is notified: activates budget code for new cost center.
6. Analytics Module refreshes org chart KPIs.
7. Portal navigation updated to reflect new org unit; team notified via Communications Module.

Alternative / Exception Flows

- Budget code not yet assigned: Finance Manager receives task to allocate before unit activation.
- Parent unit is inactive: request rejected with reason; initiator notified.

Postconditions

- OrgUnit record active in hierarchy
- Cost center budget code live
- admin.org_unit.created event emitted

UC-ADM-02 Publish an Internal Policy

Actors	Trigger	Modules Involved
Policy Owner, HR Manager (P-10) or Compliance Officer (P-11)	New regulatory requirement or internal governance decision requires a new policy	Administration (01), CMS (06), Process (29), GRC (15), Communications (07)

Preconditions

- Policy draft exists as a CMS content asset
- Approval workflow configured for the policy domain

Main Success Flow

8. Policy Owner creates AdminPolicy record linked to CMS draft document.
9. Process Module triggers policy review workflow; reviewers assigned.
10. Reviewers comment, request changes, or approve via Work Module tasks.
11. Once quorum of approvers achieved, Policy Owner publishes policy.
12. Policy status set to Active; CMS content asset published.
13. Administration Module registers effective date; old version archived.
14. Communications Module distributes policy notice to affected org units.
15. GRC Module updates compliance obligation register if policy is regulatory.
16. Knowledge Module creates or updates related knowledge article.

Alternative / Exception Flows

- Reviewer requests material changes: policy returned to draft; review cycle restarts.

Postconditions

- Policy active and accessible in policy library
- admin.policy.published event emitted
- Affected staff notified

3.2 Finance & Accounting

UC-FIN-01 Process a Supplier Invoice (AP)		
Actors	**Trigger**	**Modules Involved**
Accounts Payable Clerk, Finance Manager (P-09)	Supplier submits invoice for goods or services delivered	Finance (14), Supply Chain (37), MDM (23), Communications (07)

Preconditions

- Purchase order exists in Supply Chain Module
- Goods receipt has been recorded
- Supplier master record exists in MDM

Main Success Flow

17. AP Clerk creates Invoice record in Finance Module linked to PO and supplier.
18. Finance Module runs 3-way match: PO quantity/price vs GR vs invoice.
19. Match passes within tolerance: invoice auto-approved and queued for payment.
20. Finance Manager reviews payment batch; authorizes run.
21. Payment posted to General Ledger; bank file generated.
22. Supplier notified of payment via Communications Module.
23. Audit record written for payment authorization.

Alternative / Exception Flows

- 3-way match fails: discrepancy task sent to Procurement; payment held.
- Invoice exceeds delegation limit: escalated to CFO for approval.

Postconditions

- Invoice settled; journal entry posted; finance.invoice.settled event emitted

UC-FIN-02 Set and Monitor Departmental Budget

Actors	Trigger	Modules Involved
Department Manager (P-06), Finance Manager (P-09)	Annual or mid-year budget planning cycle initiated	Finance (14), Analytics (02), Administration (01), Work (40)

Preconditions

- Org unit exists in Administration Module
- Chart of accounts configured
- Finance period open

Main Success Flow

24. Finance Manager creates BudgetLine records for each cost center and account for the period.
25. Department Managers review and propose adjustments via Work Module tasks.
26. Finance Manager approves final budget; lines locked.
27. Finance Module begins tracking actuals vs budget on every journal posting.
28. Analytics Module computes budget variance in real time.
29. Automatic alert emitted when any line exceeds configured variance threshold.
30. Department Manager receives alert in portal notification center; reviews drill-down.
31. Finance Manager can reallocate budget between lines with approval workflow.

Alternative / Exception Flows

- Budget variance within tolerance: informational notification only, no action required.

Postconditions

- Budget lines set; actuals tracked; finance.budget.set event emitted

3.3 Legal Entity Management

UC-LEG-01 Register and Evaluate a New Legal Entity

Actors	Trigger	Modules Involved

Legal Officer (P-11), Company Secretary	Corporate structure change: new subsidiary or branch to be incorporated	Legal Entity (13), GRC (15), Board (04), MDM (23), Finance (14), Schedule (35)

Preconditions

- Organization is bootstrapped in UME kernel
- Jurisdiction policies loaded in Chombo policy pack engine

Main Success Flow

32. Legal Officer creates LegalEntityRecord with name, type, jurisdiction, and registration number.
33. Chombo auto-triggers compliance evaluation on registration (if configured).
34. 45 policy packs execute in parallel: each emits ComplianceAlerts and FilingRecords.
35. ChomboEvaluation returned: list of compliance alerts and filing schedule.
36. Legal Officer reviews alerts; critical alerts trigger task creation in Work Module.
37. Filing schedule registered; upcoming deadlines added to Schedule Module.
38. Board Module notified if entity creation requires board resolution.
39. MDM Module updates corporate hierarchy golden record.
40. Finance Module activated for new entity: chart of accounts and cost center created.

Alternative / Exception Flows

- Missing required fields (name, registration number): ENTITY-NAME-REQUIRED alert raised; entity saved in Incomplete status.

Postconditions

- Entity active in Chombo registry
- Filing schedule live
- chombo.entity.registered event emitted

UC-LEG-02 Monitor Statutory Filing Deadlines

Actors	Trigger	Modules Involved
Legal Officer (P-11), Company Secretary, System (automated)	Scheduled daily compliance check by Chombo subsystem	Legal Entity (13), Schedule (35), Communications (07), GRC (15), CMS (06)

Preconditions

- Entity registered with active filing schedule
- Filing lead days configured in Chombo settings

Main Success Flow

41. Chombo evaluates all entities with upcoming filings within the lead-day window.
42. chombo.filing.due events emitted for each approaching deadline.
43. Schedule Module adds filing tasks to enterprise calendar.
44. Legal Officer receives notification in portal with filing details and instructions.
45. Legal Officer prepares filing; marks FilingRecord as In Progress.
46. On completion, Legal Officer marks FilingRecord as Filed; evidence attached via CMS.
47. If deadline passes without filing: chombo.filing.overdue event emitted; escalation triggered.

48. Overdue filing escalated to CEO and GRC Officer via Communications Module.

Alternative / Exception Flows

- Filing extension granted: Legal Officer updates due date with regulator reference number.

Postconditions

- Filing record updated; compliance status current; audit record written

3.4 Human Resources

UC-HR-01 Run a Performance Review Cycle		
Actors	**Trigger**	**Modules Involved**
HR Manager (P-10), Manager (P-06), Employee (P-17)	Annual or mid-year performance review cycle initiated by HR	HR (16), Work (40), Learning (20), Analytics (02), Communications (07)

Preconditions

- All active employees have a manager assigned
- Review templates configured in Templating System
- Previous review records available

Main Success Flow

49. HR Manager creates review cycle; selects period and template; assigns to all active employees.
50. Employees receive notification: self-assessment tasks created in Work Module.
51. Employees complete self-assessments via portal; submitted to manager.
52. Managers receive review tasks; access employee self-assessments and previous OKR progress.
53. Managers complete assessments; assign ratings; write development comments.
54. HR Manager reviews rating distribution; flags anomalies for calibration.
55. Calibration session held (recorded as board/committee meeting if required).
56. Reviews finalized; development plans auto-created in Work Module.
57. Learning Module assigned learning paths based on development plan gaps.
58. Employees receive completed review; acknowledge via portal.

Alternative / Exception Flows

- Manager misses review deadline: escalation task sent to HR Manager.
- Rating calibration dispute: HR Manager mediates and updates final rating.

Postconditions

- All reviews completed; hr.review.completed event emitted; development plans active

3.5 Governance, Risk & Compliance

UC-GRC-01 Register and Treat an Enterprise Risk

Actors	Trigger	Modules Involved
Risk Manager (P-12), Risk Owner, Executive Sponsor	New risk identified through incident, environmental scan, or audit finding	Risk (33), GRC (15), Board (04), Analytics (02), Work (40)

Preconditions

- Risk appetite framework defined in Risk Module
- Risk categories and scoring matrix configured

Main Success Flow

59. Risk Manager creates Risk record: title, category, description, initial scores.
60. Risk Module computes inherent risk score from likelihood × impact on configured matrix.
61. Risk Manager assigns owner and executive sponsor; treatment type selected.
62. Risk Owner creates TreatmentPlan: actions, controls, target residual score, due dates.
63. Controls linked from GRC control library; design assessment completed.
64. KRIs defined for the risk; thresholds set; KRI monitoring activated.
65. Risk presented at next risk committee meeting via Board Module.
66. Risk Manager updates residual score as controls are implemented.
67. KRI monitor emits alert if risk.kri.amber_threshold or risk.kri.red_threshold breached.

Alternative / Exception Flows

- Risk score exceeds appetite threshold: auto-escalation to CEO and Risk Committee.
- Treatment plan due date missed: overdue task escalation to Risk Owner and sponsor.

Postconditions

- Risk registered; KRIs active; treatment plan in progress

3.6 Sales & Customer Relationship Management

UC-SAL-01 Manage a Sales Opportunity from Lead to Close

Actors	Trigger	Modules Involved
Sales Representative (P-13), Sales Manager, Finance Manager (P-09)	Marketing-qualified lead passed from Marketing Module, or direct outbound prospect identified	Sales (34), CRM (08), Marketing (22), Finance (14), Legal Entity (13), Templates (39)

Preconditions

- Stakeholder record exists or created in CRM
- Product catalog active in Product Module
- Sales stages configured

Main Success Flow

68. Lead arrives from Marketing Module campaign (soko.campaign.evaluated triggers CRM lead).
69. Sales Rep creates SalesOpportunity linked to CRM account; initial qualification.
70. Sales Rep logs discovery call activity; outcome and next steps recorded.
71. Sales Rep advances opportunity through pipeline stages: Qualify → Propose → Negotiate.
72. Finance Module provides product pricing from price list for proposal.
73. Sales Rep produces and sends proposal; activity logged against opportunity.
74. Negotiation complete: opportunity advanced to Close stage.
75. Contract generated from Templates Module; legal reviews contract.
76. Contract signed: Contract record created in CRM; opportunity marked Closed Won.
77. Finance Module triggers invoice generation for first billing milestone.
78. Sales commission computed by Sales Module for the period.

Alternative / Exception Flows

- Prospect selects competitor: Closed Lost recorded with reason; loss analysis report generated.
- Deal exceeds delegation authority: CFO approval required before contract signed.

Postconditions

- Opportunity closed; CRM account updated; sales.opportunity.closed_won emitted; invoice issued

3.7 IT & Security

UC-IT-01 Manage an IT Incident to Resolution

Actors	Trigger	Modules Involved
IT Administrator (P-16), Affected User (P-17), IT Manager	User reports service disruption or automated monitoring alert detected	IT (18), Security (36), Knowledge (19), Communications (07), Work (40)

Preconditions

- Affected system registered as CI in CMDB
- SLA definitions configured
- On-call rotation assigned

Main Success Flow

79. User submits incident via portal service request or alert auto-creates incident.
80. IT Module classifies severity; starts SLA timer; assigns to on-call engineer.
81. IT Admin reviews incident; identifies affected CIs via CMDB dependency map.
82. IT Admin applies initial workaround; updates affected users via Communications Module.
83. Root cause investigated; if change required: Change Request created.
84. Change advisory board reviews change request; approves for implementation.
85. IT Admin implements fix; validates service restoration; closes incident.
86. Incident summary sent to affected user; satisfaction survey issued.
87. Post-incident review initiated if P1/P2; lessons captured in Knowledge Module.

Alternative / Exception Flows

- SLA breach imminent: escalation notification to IT Manager.
- Change rejected by CAB: alternative workaround sought or incident re-scoped.

Postconditions

- Incident closed; CMDB updated; SLA compliance recorded; lessons captured if P1/P2

3.8 Marketing

UC-MKT-01 Launch and Evaluate a Marketing Campaign		
Actors	**Trigger**	**Modules Involved**
Marketing Manager (P-14), Creative Team, Analytics Team	Campaign brief approved as part of quarterly marketing plan	Marketing / Soko (22), Analytics (02), Design (09), Communications (07), Sales (34)

Preconditions

- Campaign audience persona defined in Soko
- Budget allocated in Finance Module
- Creative assets available in CMS/Design Module

Main Success Flow

88. Marketing Manager registers MarketingCampaign in Soko with channels, budget, and timeline.
89. Soko evaluates campaign: 70 strategy packs run in parallel generating insights and activation actions.
90. Marketing Manager reviews insights; selects activation actions to implement.
91. Creative assets linked from Design Module brand asset library.
92. Campaign launched via Communications Module across configured channels.
93. Market signals ingested in real-time as campaign runs: click, open, conversion signals.
94. Soko re-evaluates campaign with accumulated signals; updated insights generated.
95. Analytics Module refreshes campaign performance dashboard: CAC, conversion, ROI.
96. Marketing Manager applies optimization actions based on updated insights.
97. Campaign concludes: final performance report generated; distributed to CMO and CEO.

Alternative / Exception Flows

- Budget threshold exceeded: alert emitted; campaign paused pending approval.
- Campaign performance below threshold after 7 days: optimization escalation to Marketing Manager.

Postconditions

- Campaign completed; performance report published; leads transferred to Sales Module

3.9 Supply Chain & Logistics

UC-SC-01 Respond to an Inventory Stockout Risk

Actors	Trigger	Modules Involved
Inventory Manager, Procurement Team, Supply Chain Module (automated)	sc.inventory.reorder_triggered event: stock level at or below reorder point	Supply Chain (37), Finance (14), Analytics (02)

Preconditions

- Item master exists with reorder point and safety stock configured
- Approved supplier list available in Supply Chain Module

Main Success Flow

98. Supply Chain Module detects item at reorder point; emits sc.inventory.reorder_triggered.
99. Module auto-creates purchase requisition for replenishment quantity.
100. Demand Planner reviews: confirms quantity based on demand forecast.
101. Procurement Team selects from approved supplier list; creates PO.
102. PO approved by manager (within delegation limit) and issued to supplier.
103. Supplier acknowledges PO; expected delivery date confirmed.
104. Shipment tracked; sc.shipment.delayed event emitted if ETA slips.
105. Goods received; GR recorded; inventory updated; stockout risk cleared.
106. Analytics Module updates inventory turns and fill rate KPIs.

Alternative / Exception Flows

- All approved suppliers cannot meet lead time: emergency sourcing escalation.
- Received goods fail quality inspection: GR marked partial; supplier NCR raised.

Postconditions

- Inventory replenished; reorder point reset; stockout risk resolved

3.10 Knowledge & Learning

UC-KNW-01 Capture and Publish a Lessons Learned Article

Actors	Trigger	Modules Involved
Project Manager (P-07), Knowledge Manager, Subject Matter Expert	Project closure or post-incident review completed	Knowledge (19), CMS (06), Templates (39), Learning (20), Portal (26)

Preconditions

- Project record exists in Portfolio Module
- CMS content workflow configured for knowledge articles

Main Success Flow

107. Project Manager submits lessons learned form via portal.
108. Knowledge Module creates LessonLearned record; links to source project.
109. Knowledge Manager reviews entry; identifies SME for article authorship.
110. SME drafts KnowledgeArticle using appropriate template from Templates Module.
111. Article enters peer review workflow via Process Module.
112. Reviewer approves; Knowledge Manager publishes article.
113. Search index updated; article discoverable via portal global search.
114. Related articles auto-linked via taxonomy and tag similarity.
115. Learning Module links article as reference material in relevant learning paths.

Alternative / Exception Flows

- Reviewer requests substantive changes: article returned to draft; revision cycle restarts.

Postconditions

- Knowledge article published; knowledge.article.published emitted; searchable via portal

3.11 Portfolio & Project Management

UC-PPM-01 Initiate a New Project in the Portfolio

Actors	Trigger	Modules Involved
Project Sponsor, Project Manager (P-07), Portfolio Manager	Business case approved by investment committee or board	Portfolio (27), Strategy (21), Work (40), Risk (33), HR (16), Finance (14), Schedule (35)

Preconditions

- Strategic objective exists in Strategy Module
- Business case approved in Investment Module
- Budget allocated in Finance Module

Main Success Flow

116. Portfolio Manager creates PortfolioItem linked to strategic pillar and approved business case.
117. Project Manager creates Project record; linked to portfolio item.
118. Project Manager decomposes scope into WBS work packages.
119. Milestones scheduled; linked to enterprise calendar via Schedule Module.
120. RAID log initialized; initial risks loaded from business case risk section.
121. Project risks with high residual scores escalated to Risk Module.
122. Resource demand entered; HR Module checked for capacity conflicts.
123. Project kickoff meeting scheduled; team notified via Communications Module.
124. Work items created in Work Module for first sprint/phase tasks.
125. Portfolio dashboard activated; project health tracking begins.

Alternative / Exception Flows

- Resource conflict detected: Portfolio Manager facilitates cross-project prioritization.

- Scope change requested mid-project: change control workflow initiated; budget and schedule re-baselined if approved.

Postconditions

- Project live in portfolio; schedule active; resources confirmed; monitoring live

3.12 Board Management

UC-BRD-01 Prepare and Run a Board Meeting		
Actors	**Trigger**	**Modules Involved**
Company Secretary, Board Members, CEO (P-01)	Scheduled board meeting date approaches (configurable lead days)	Board (04), Finance (14), Strategy (21), CMS (06), Communications (07), Templates (39)

Preconditions

- Board meeting scheduled in Board Module
- Board pack lead-day configuration set
- Directors registered with contact details

Main Success Flow

126. Board Module triggers board pack assembly lead_days before meeting.
127. Company Secretary confirms agenda with CEO; AgendaItems created.
128. Committee reports, financial statements, and strategy updates pulled from respective modules.
129. Board pack assembled from Templates Module; branded and compiled.
130. Board pack distributed to all directors via Communications Module; delivery logged.
131. Company Secretary confirms quorum on meeting day; QuorumStatus recorded.
132. Meeting runs; Company Secretary records minutes in real time.
133. Resolutions proposed, voted on, and recorded with vote outcome.
134. Post-meeting: minutes distributed for review and approval at next meeting.
135. Approved minutes published in CMS; resolutions logged in statutory register.

Alternative / Exception Flows

- Quorum not achieved: meeting lapsed; board.quorum.failed event emitted; meeting rescheduled.
- Director declares conflict of interest: declaration logged; director excluded from relevant agenda item.

Postconditions

- Resolutions formally recorded; minutes approved and published; board.resolution.passed emitted

4. End-to-End User Journey Maps

User journeys describe the experience of a persona interacting with the UME OS over a continuous sequence of tasks. These journeys cut across multiple sessions and modules.

Journey J-01: The CEO's Monday Morning

Persona: CEO (P-01). Context: Start-of-week executive review routine.

Time	Action	Module	What CEO Sees
07:45	Opens UME portal on mobile	Portal (26)	Personalized executive dashboard: 5 KPI tiles, 3 unread critical notifications
07:47	Reviews OKR confidence scores	Strategy (21) via Analytics (02)	2 OKRs flagged at-risk (confidence <40%); owners listed; trend charts
07:50	Taps risk notification: KRI breached red threshold overnight	Risk (33)	Risk detail card: Cybersecurity breach risk; KRI value, treatment status, owner contact
07:53	Reviews financial snapshot tile	Finance (14) via Analytics (02)	MTD revenue vs budget: -8% variance; top line and EBITDA; quick drill-down to Finance module
08:00	Checks board resolution tracker	Board (04)	3 open resolutions pending owner action; 1 overdue flagged in red
08:05	Reviews strategic initiative health	Strategy (21)	Portfolio of 12 initiatives: 8 green, 3 amber, 1 red (milestone missed)
08:10	Sends nudge to initiative owner via portal message	Communications (07)	Message sent; logged as CRM interaction with executive stakeholder
08:15	Closes portal; attends first meeting with full situational awareness	—	CEO informed without any manual report preparation or email triage

Journey J-02: The New Employee's First Week

Persona: New Employee (P-17). Context: First week at the organization.

Day	Action	Module	Experience
Day 1 AM	Receives welcome email with portal login link	Communications (07)	Branded welcome email from template; single-click portal access
Day 1 AM	Logs into portal; sees personalized onboarding checklist	Portal (26) + Work (40)	Onboarding tasks listed with priorities; 0/12 complete indicator
Day 1 PM	Completes IT account setup and mandatory security training	IT (18) + Learning (20)	IT tasks auto-complete on system provisioning; security course assigned

Day 2	Reads team handbook and org chart	Knowledge (19) + Admin (01)	Handbook article found via global search; org chart interactive in portal
Day 2	Submits laptop equipment issue via IT service request	IT (18)	Service request created; IT admin notified; SLA timer starts; status trackable
Day 3	Enrolls in onboarding learning path: 5 courses assigned	Learning (20)	Progress bar shows 0/5 modules; first course started; CPD points awarded on completion
Day 3	Meets manager for 30-day goal setting; OKRs created	Strategy (21) + Work (40)	OKRs appear on employee portal dashboard; work items created for first sprint
Day 5	Completes all day-1 mandatory tasks; receives notification	Learning (20) + Portal (26)	Completion badge in portal; onboarding progress 8/12; remaining tasks listed
Week 2	Independently navigates portal; uses global search daily	Portal (26) + Knowledge (19)	Employee self-sufficient; average search-to-answer under 30 seconds

Journey J-03: The Project Manager's Sprint Cycle

Persona: Project Manager (P-07). Context: Managing a 2-week delivery sprint within an active portfolio project.

When	Action	Module	Outcome
Sprint Start	Reviews WBS; creates sprint backlog from portfolio work packages	Portfolio (27) + Work (40)	Sprint created; work items scoped and estimated; team notified
Sprint Start	Confirms resource availability; checks calendar for conflicts	HR (16) + Schedule (35)	No capacity conflicts; sprint plan confirmed; kickoff event scheduled
Daily	Reviews kanban board; updates item statuses after stand-up	Work (40)	WIP visible; blockers flagged; velocity metric updating in real time
Day 4	Blocked item: dependency on IT change not yet approved	Work (40) + IT (18)	Dependency link created; IT Admin receives blocker notification; escalation path visible
Day 7	Mid-sprint: budget actuals reviewed against plan	Finance (14)	Budget burn-rate on track; no variance alert
Day 9	New requirement identified: submits change request	Requirements (32) + Portfolio (27)	Change request impact analysis run; scope change approved; sprint adjusted
Day 12	Sprint retrospective run in Teams Module	Teams (38)	Retro items recorded; 3 action items created in Work Module with owners
Sprint End	Velocity recorded; burndown chart generated	Work (40) + Analytics (02)	Sprint metrics published; portfolio health dashboard updated

Sprint End	Status report generated from template; sent to stakeholders	Templates (39) + Communications (07)	Branded status report distributed; delivery confirmed in audit trail

Journey J-04: The Finance Manager's Budget Cycle

Persona: Finance Manager (P-09). Context: Annual budget setting and ongoing budget monitoring.

Phase	Action	Module	Outcome
Planning	Downloads budget template pre-populated with prior year actuals	Finance (14) + Templates (39)	Template rendered with actuals; sent to department managers for input
Planning	Collects departmental budget submissions via process workflow	Process (29) + Work (40)	Budget submission tasks assigned to managers; due date tracked
Planning	Consolidates submissions; identifies gaps vs strategic plan	Finance (14) + Strategy (21)	Consolidated budget view; OKR-linked budget lines visible
Approval	Submits budget to CFO and board for approval	Finance (14) + Board (04)	Board agenda item created; budget pack assembled and distributed
Approval	Board approves budget; resolution recorded	Board (04)	Budget lines locked; Finance Module activated for tracking
Month 1	First month closes: actuals vs budget variance computed	Finance (14)	Variance report auto-generated; dept managers notified of their line performance
Month 2	Sales revenue 12% below budget: alert triggered	Finance (14) + Analytics (02)	CFO dashboard flagged; drill-down to sales pipeline shows pipeline gap
Month 3	Reforecast initiated: rolling forecast updated with new assumptions	Finance (14)	Updated forecast lines committed; board notified of reforecast vs budget
Year End	Full year results vs budget: variance analysis report	Finance (14) + Analytics (02)	Comprehensive variance report; input to next annual planning cycle

5. System Administration & Platform Use Cases

These use cases cover the UME system operator persona (P-18) managing the UME OS platform itself.

UC-SYS-01 Onboard and Configure a New Organization Module

Actors	Trigger	Modules Involved
UME System Operator (P-18), IT Administrator (P-16)	Organization decision to activate a previously inactive module	Enterprise Management (12), IT (18), Security (36), Backup (03)

Preconditions

- Module binary or crate available in workspace
- Module dependencies already active
- RBAC roles defined for module domain

Main Success Flow

136. System Operator opens Enterprise Management Module: views current module registry.
137. Operator proposes config change: enables new module in KernelConfig.
138. Config change approval workflow triggered; second operator approves.
139. Kernel applies config change: module initialized and registered in ModuleRegistry.
140. Module boots: emits module.registered event; health check passes.
141. System Operator verifies module health in system overview dashboard.
142. RBAC roles and permissions assigned to relevant user groups.
143. Portal updated: new module navigation link appears for authorized users.
144. Operator verifies first successful operation; closes change record.

Alternative / Exception Flows

- Module dependency not satisfied: boot fails with dependency error; system overview shows Error state.
- Health check fails: supervisor applies restart policy; operator notified.

Postconditions

- Module active and healthy; users can access; audit record of activation created

UC-SYS-02 Schedule and Verify a System Backup

Actors	Trigger	Modules Involved
UME System Operator (P-18)	Initial system setup or change to backup schedule required	Backup (03), Security (36), Enterprise Management (12)

Preconditions

- Backup destination storage driver registered
- Encryption keys available via Security Module vault

Main Success Flow

145. System Operator creates BackupJob in Backup Module: scope, type, schedule, retention, destination.
146. Backup Module schedules job on configured cron expression.
147. At scheduled time: backup job runs; all module snapshots captured.
148. Storage driver writes encrypted backup artifact to configured destination.
149. BackupManifest created with artifact path, checksum, and chain reference.
150. Integrity check runs automatically: checksum verified post-write.
151. backup.job.completed event emitted; operator receives portal notification.
152. Operator optionally runs manual DR test: restore-to-sandbox executed.
153. Restore validation passes: backup confirmed healthy and restorable.

Alternative / Exception Flows

- Checksum verification fails: backup.integrity.failed event emitted; operator alerted immediately; backup flagged invalid.
- DR test fails: backup.dr_test.failed event emitted; escalation to IT management.

Postconditions

- BackupManifest recorded; backup verified; schedule active for next run

UC-SYS-03 Manage RBAC Roles and Permission Assignments

Actors	Trigger	Modules Involved
UME System Operator (P-18), HR Manager (P-10)	New employee joins, role changes, or new module activated requiring permission setup	Security (36), HR (16), Enterprise Management (12)

Preconditions

- Employee record exists in HR Module
- Module permissions declared in ModuleDescriptor

Main Success Flow

154. System Operator reviews role definitions in Security Module RBAC configuration.
155. Operator creates or updates Role with required permissions for module domain.
156. HR Manager identifies employees requiring the role based on job function.
157. Operator assigns Role to Subject (employee) in RbacEngine.
158. RBAC can() check now returns true for assigned permissions.
159. Employee accesses module; permission check passes; access logged in audit.
160. Access review campaign scheduled via Security Module (periodic certification).
161. During access review: manager certifies or revokes each assignment.
162. Revoked assignments removed; security.access_review.completed event emitted.

Alternative / Exception Flows

- Employee attempts access without required permission: security.rbac.denied event emitted; audit record written.
- Role assignment above privilege level: second approver required (separation of duties).

Postconditions

- RBAC assignments current; audit trail complete; access review scheduled

6. Custom Module Extension Use Cases

These use cases demonstrate how development teams extend the UME OS with custom modules.

UC-EXT-01 Develop and Deploy a Custom Organization Module

Actors	Trigger	Modules Involved
Software Engineer (P-15), UME System Operator (P-18)	Organization requires domain-specific functionality not covered by built-in modules	Enterprise Management (12), Security (36), All modules (via event bus integration)

Preconditions

- UME module SDK installed
- ume_core available as dependency
- Custom module ID namespace reserved with operator

Main Success Flow

163. Engineer runs module scaffold generator: tools/generate_custom_module produces boilerplate crate.
164. Engineer implements KernelModule trait methods: id, domain, state, start, stop, health_check, describe.
165. Engineer declares module dependencies and permissions in ModuleDescriptor.
166. Engineer implements domain logic with RBAC guards on all mutating operations.
167. Engineer emits domain events on all state-changing calls using kernel event bus.
168. Engineer writes audit records for all operations via LogAuditManager.
169. Unit tests and contract tests written and pass CI.
170. System Operator reviews module manifest; approves deployment in sandbox mode.
171. Sandbox validation passes; Operator promotes module to active registration.
172. Module boots healthy; portal widget registered; permissions assigned to relevant roles.

Alternative / Exception Flows

- Module fails health check on first boot: supervisor applies restart policy; engineer investigates logs.

- Missing declared dependency: boot rejected; engineer resolves dependency before re-attempting.

Postconditions

- Custom module live; healthcheck passing; users provisioned with permissions; audit trail active